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# Course: Bioinformatics - ENBC 455
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# Problem 1
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```
with open('hw3_1.txt') as file:
    months, litter_mul = file.readline().split()
    adults = 1
    babies = 0
    newborns = 0

    for i in range(3, int(months) + 1):
        adults += babies
        babies = newborns
        newborns = adults * int(litter_mul)

    print(adults + babies + newborns, "\n")
```

```
from Bio.Seq import Seq
```

```
# Problem 2
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```
with open('hw3_2.txt') as file:
    # Filter all sequences from the file into a list
    seqs = []
    temp = ""
    for line in file.readlines():
        if '>' not in line:
            temp += line.strip()
        else:
            if temp:
                seqs.append(temp)
            temp = ""
    if temp:
        seqs.append(temp)

    # Remove all of the introns within the original sequence
    raw_seq = seqs.pop(0)
    for seq in seqs:
        raw_seq = raw_seq.replace(seq, "")

    # Translate sequence into protein string
    dna_seq = Seq(raw_seq)
    protein_string = dna_seq.translate()
    print(protein_string, "\n")
```

```
# Problem 3
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```
with open('hw3_3.txt') as file:
    # Filter all sequences from the file into a list
    seqs = []
```

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40
41     # Translate sequence into protein string
42     dna_seq = Seq(raw_seq)
43     protein_string = dna_seq.translate()
44     print(protein_string, "\n")
45
46 # Problem 3
47 with open('hw3_3.txt') as file:
48     # Filter all sequences from the file into a list
49     seqs = []
50     temp = ""
51     for line in file.readlines():
52         if '>' not in line:
53             temp += line.strip()
54         else:
55             if temp:
56                 seqs.append(temp)
57             temp = ""
58     if temp:
59         seqs.append(temp)
60
61     # Identify subsequence indexes
62     target = 0
63     for i in range(len(seqs[0])):
64         if target == len(seqs[1]):
65             break
66
67         if seqs[0][i] == seqs[1][target]:
68             print(i + 1, end=" ")
69             target += 1
70     print("\n")
71
72 # Problem 4
73 with open('hw3_4.txt') as file:
74     # Filter all sequences from the file into a list
75     seqs = []
76     temp = ""
77     for line in file.readlines():
78         if '>' not in line:
79             temp += line.strip()
80         else:
81             if temp:
82                 seqs.append(temp)
83             temp = ""
84     if temp:
85         seqs.append(temp)
86
87     # Identify shortest sequence
88     seqs.sort(key=len)
89     shortest_seq = seqs[0]
90     other_seqs = seqs[1:]
91     cond = False
92
93     # Loops through possible lengths of substring backwards to have the
94     # longest common subsequence guaranteed later on
95     for length in range(len(shortest_seq), 0, -1):
96         # Acts as a sliding window
97         for i in range(len(shortest_seq) - length + 1):
98             candidate = shortest_seq[i:i+length]
99
100             # Checks if current candidate sequence exists in all other
101             # sequences
102             cond = True
103             for seq in other_seqs:
104                 if candidate not in seq:
105                     cond = False
106                     break
107             # If condition is met and the longest common subsequence is found
108             # in all of the sequences, then it is printed out
109             if cond:
110                 print(candidate)
111                 break
112     if cond:
113         break

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MGAAPGRLSYKLADCLHPSSAVKSRTNTLQGTPIAVLSSQLRAYPVNMCLPISYACIPESAALALRRLYTSMEREHGKEGVPAGQGTFGVRLVPYNKDVRFLVVRPLRFEARRRRRPL
DLTLGQNQASPNGESYGTFTATKSL SPLHSLGQGRLILTTRKSYFQSDLQCARIVRSTHVVYVSSSNRAWP*

2 3 5 8 9 10 11 12 15 19 20 21 22 26 29 33 36 45 46 48 49 63 70 71 74 76 78 80 89 91 94 95 96 105 109 111 113 115 129 1
32 136 138 147 150 154 155 157 159 160 162 164 169 174 178 181 190 195 196 199 202 210 212 226 230 237

CAAAAGCGACGTAAGGTGCCTAAGAGCGATCCGTCCCGTGAAGTCACAGAATCGCGATG